Damage Predicts Recovery: When Calibration Data Matters in Compressing Financial LLMs

Junyi Ye
School of Computing
Montclair State University
Montclair, New Jersey, USA
yej@montclair.edu

Mengjia Yu
School of Computing
Montclair State University
Montclair, New Jersey, USA
yum1@montclair.edu

Debapriya Hazra
School of Computing
Montclair State University
Montclair, New Jersey, USA
hazrad@montclair.edu

Guiling Wang
Ying Wu College of Computing
New Jersey Institute of Technology
Newark, New Jersey, USA
gwang@njit.edu

Abstract

Post-training quantization and pruning rely on a small calibration corpus. Whether specialized domains such as finance require domain-matched calibration data remains unsettled. We argue that the answer depends on the task-level damage caused by compression rather than on domain mismatch. If compression preserves the target capability, changing the calibration corpus has little effect. If compression causes large losses, task-formatted calibration can recover part of the loss. We test this hypothesis across two model families, six compression configurations, three token-matched calibration corpora, and ten financial classification and numerical question-answering tasks. The results support this hypothesis. Quantization largely preserves task performance, and calibration choice has little effect in this case. Pruning reduces numerical QA accuracy by up to 41 points. In these damaged settings, another generic corpus does not help, while FinMix, a mixture of financial task examples, recovers part of the loss (roughly 20–60% of numerical-QA damage under pruning). The link between damage and recovery holds across model families and scales. These findings support a practical rule. Measure task-specific compression damage first, and construct specialized calibration data only when the damage is large.

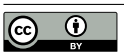
Code Availability: https://github.com/JunyiYe/finance-ptc-calibration

CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Neural networks**; • **Applied computing** → *Economics*.

Keywords

post-training compression, quantization, pruning, calibration data, financial NLP, numerical reasoning



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1 Introduction

Post-training compression makes large language models cheaper to deploy, but many widely used quantization and pruning methods depend on a small calibration corpus. In practice, this corpus is often generic text such as WikiText or C4, even when the compressed model will be used in a specialized domain [3, 4, 7, 15, 21]. Whether domain- and task-matched calibration improves downstream performance remains unsettled: prior results range from substantial gains to little change or even degradation [6, 17, 18]. Consequently, practitioners lack a clear rule for deciding when specialized calibration data is worth constructing.

We argue that this question should not be answered from domain mismatch alone. The relevant quantity is the task-specific *damage* caused by a particular model and compression method under generic calibration. If compression already preserves the target capability, changing the corpus leaves little room for improvement. If it severely degrades that capability, task-formatted calibration may recover part of the loss. Financial NLP provides a revealing test case because its tasks range from label-based classification to numerical question answering over reports and tables, allowing compression to preserve one capability while severely degrading another.

We test this hypothesis in a controlled study of six compression configurations, three calibration corpora, and ten financial tasks. The main experiments use Llama-3.1-8B, while Qwen3-8B evaluates transfer to a second model family and Llama-3.1-70B evaluates scale. For each calibration-dependent method, we compare WikiText and C4 with FinMix, a token-matched mixture of financial training examples in the target-task format. Every compressed model is compared with its BF16 reference on the same test examples using paired statistical tests; free-form numerical answers are evaluated with a human-validated LLM judge.

Quantization usually leaves performance close to BF16, so calibration choice has little effect. Pruning causes much larger losses, especially on numerical QA, and generic-text perplexity does not reliably reflect them. In these damaged settings, FinMix recovers performance more consistently than swapping in another generic corpus. Damage and recovery are strongly associated across Llama-3.1-8B and Qwen3-8B (Spearman $\rho = 0.74$ over 90 observations), and the smaller gains at 70B follow from its smaller damage. Calibration effort should therefore be matched to measured compression damage, not assigned solely because the deployment domain is financial.

Our contributions are threefold:

- **Capability-selective damage.** We show that compression, especially pruning, disproportionately damages financial numerical QA, while classification is often preserved and generic perplexity can miss the resulting loss.
- **Damage-dependent recovery.** We show that financial, task-formatted calibration is most effective when generic calibration has caused substantial task-level damage. A second generic-corpus control and repeated calibration draws confirm that this recovery is not merely caused by changing the calibration sample. Paired error analysis further shows that FinMix primarily restores answers that the uncompressed model could already produce.
- **A generalizable decision rule.** We show that task-level damage is a strong predictor of the benefit of changing the calibration corpus across the model families and scales tested, supporting a practical damage audit before investing in specialized calibration data.

2 Related Work

Calibration data in post-training compression. Calibration-dependent methods such as GPTQ, AWQ, SmoothQuant, SparseGPT, and Wanda typically use a small sample of generic or pretraining-like text [3, 4, 7, 15, 21]. Their original studies focus on the compression algorithms rather than comparing calibration corpora. Williams and Aletras [17] later showed that the corpus can change downstream performance, particularly under pruning. Subsequent work replaces random generic text with data closer to the pretraining distribution [6], model-generated text [18], or mixtures of generic and domain- or task-specific examples. Williams et al. [19] propose a calibration method that improves the in-domain performance of compressed models in domains such as biomedicine and law, and Wang et al. [16] report that task-only calibration can hurt mixed-precision quantization while added general-domain data helps. We do not propose a new calibration method. We ask when specialized calibration data is worth building at all.

Evaluating compressed LLMs. Perplexity is commonly used to evaluate compression, but it does not show whether individual capabilities remain intact. Jaiswal et al. [5] find that pruning can leave perplexity nearly unchanged while substantially degrading knowledge-intensive tasks. In their evaluation, quantization is more robust. Liu et al. [8] further show that quantization effects depend on both precision and capability: in-context learning, chain-of-thought reasoning, and instruction following largely survive at 4 bits but deteriorate sharply at 2 bits. These findings motivate task-level evaluation rather than reliance on perplexity alone.

Financial LLMs and benchmarks. BloombergGPT [20], FinGPT [24], and PIXIU [23] develop financial models through domain pretraining or instruction tuning. PIXIU and FinBen [22] also consolidate benchmarks covering financial classification, information extraction, and numerical reasoning. These studies focus on building and evaluating financial LLMs. They do not examine how post-training quantization and pruning affect these capabilities under different calibration corpora.

3 Experimental Setup

3.1 Models and Compression Configurations

The main experiments use Llama-3.1-8B. Qwen3-8B tests whether the findings transfer to a second model family, and Llama-3.1-70B tests whether they remain at a larger scale. We use the Instruct version of each model and compare every compressed model with its corresponding BF16 checkpoint.

We evaluate six compression configurations produced with `llm-compressor`[1]. RTN, GPTQ [4], and AWQ [7] use W4A16 quantization, while SmoothQuant [21] uses W8A8. The pruning configurations are SparseGPT [3] at 50% unstructured sparsity and Wanda [15] at 4:8 semi-structured sparsity. RTN does not use calibration data and therefore serves as a calibration-free reference. Each of the other five configurations is evaluated with the three calibration corpora introduced next. GPTQ, SparseGPT, and Wanda use corpus statistics to determine how weights are quantized or removed, whereas AWQ and SmoothQuant use them to set scaling factors.

3.2 Calibration Corpora

Table 1 summarizes the three calibration corpora and their roles. The comparison distinguishes a generic-corpus swap from a switch to financial, task-formatted calibration data. WikiText is the generic reference in the main damage analysis and follows the original GPTQ setup. C4 provides a second generic corpus and follows the original AWQ setup. Comparing it with WikiText shows whether performance changes after replacing one generic corpus with another. FinMix then tests the combined benefit of financial content and task-style input format.

FinMix is constructed from the training splits of the ten public financial benchmarks listed in Table 2. We give each task the same token budget so that no single benchmark dominates the mix. Each training example is converted into the same input format used to evaluate its task. We remove any example that duplicates an item in a test set.

All five calibration-based methods consume each corpus in full. WikiText and C4 each contain 128 sequences truncated to 2048 tokens, for a total of 262,144 tokens. FinMix contains 2,042 shorter examples whose lengths vary by task and total approximately 260k tokens. The corpora therefore match the amount of calibration text, although they differ in the number and length of examples.

Prior work finds diminishing returns from adding calibration examples at this scale and commonly uses 128 long sequences

[1] https://github.com/vllm-project/llm-compressor

Table 1: Calibration corpora used in the experiments. All calibration-based methods consume each corpus in full.

Corpus	Content	Role	Seqs	Tokens
WikiText	encyclopedia text	reference	128	262k
C4	web text	generic control	128	262k
FinMix	financial tasks	matched corpus	2,042	260k

Table 2: Evaluation tasks. n = canonical test-set size. Balance = class distribution (%); for FinRED (29 relation types) we report only the largest class share, and the three free-form QA tasks have no fixed label set (–). Num. = degree of numerical dependence. Classification tasks report weighted F1, except FinCausal, which reports macro-F1. QA tasks report accuracy (judge-verified for the generative tasks, marked †).

Task	Type	n	Balance	Num.	Metric
FPB [9]	3-cls sentiment	970	59/29/12	none	wF1
FOMC [11]	3-cls stance	496	49/26/25	none	wF1
Headlines [13]	binary attribute	1500	67/33	low	wF1
FinRED [12]	29-cls relation	1500	max 13	none	wF1
FinCausal [10]	2-cls causality	1500	94/6	low	mF1
BigData22 [14]	2-cls movement	1472	55/45	low	wF1
CFA [23]	3-way exam QA	1032	35/34/31	medium	Acc
FinQA [1]	numerical QA	1147	–	high	Acc†
ConvFinQA [2]	conv. numerical QA	1490	–	high	Acc†
TAT-QA [25]	table+text QA	1500	–	high	Acc†

[3, 15, 17]. The larger number of FinMix examples should therefore not be read as a larger calibration budget: it reflects the shorter length of task examples, while the total number of tokens remains matched. The FinMix comparison intentionally changes both financial content and input format. We interpret its effect as their practical combination rather than as a pure domain effect, since domain-specific examples normally appear in the format of the target task.

3.3 Evaluation Tasks

Table 2 reports the size, label balance, numerical dependence, and metric for each task. All ten come from public financial benchmarks. We use the `flare` versions distributed with PIXIU [23] where available and evaluate on the official test splits. The selected tasks span a broad range of financial language understanding and reasoning capabilities, including sentiment and stance classification, relation and causal understanding, directional prediction, question answering, and multi-step numerical reasoning.

Six tasks use classification labels. FPB identifies the sentiment of a financial sentence, and FOMC identifies the policy stance of a central-bank statement. Headlines pairs a gold-market headline with one of nine yes/no questions about attributes such as price direction, time, events, or asset comparisons. FinRED identifies the relation between two financial entities in context. FinCausal determines whether a financial causal relation is expressed. BigData22 predicts the next-day direction of a stock from tweets and recent price history.

Four tasks use question answering. CFA contains multiple-choice questions about financial concepts and analysis. FinQA requires the model to find values in financial reports and combine them through multiple calculation steps. ConvFinQA adds conversational follow-up questions whose answers can depend on earlier turns. TAT-QA asks for calculated or extracted answers from evidence presented jointly as a table and accompanying text.

3.4 Evaluation and Scoring

Every configuration of a task is scored on the same test examples and the same prompts, and an automated check verifies this before scoring. Decoding is greedy throughout, chat templates are fixed within each model family, and Qwen3-8B runs with thinking mode disabled.

Classification tasks are scored from label log-probabilities rather than from generated text, so a degraded model cannot lose points for formatting alone. Headlines compares the single tokens Yes and No. FPB compares the three sentiment words, which tokenize to matched lengths. The labels of FOMC, FinRED, FinCausal, and BigData22 tokenize to unequal lengths, which direct scoring would reward, so their choices are mapped to single-token option letters, the mapping is rotated through every position, and the scores are averaged; the rotation removes both the length advantage and any preference for a particular letter or position. Each scorer verifies its tokenization and option mapping at startup and stops rather than mis-score. CFA is the exception: the model reasons freely and its final choice among the three options is extracted, and a response with no extractable choice counts as incorrect.

CFA and the three free-form QA tasks generate up to 2,048 tokens, greedily. The QA tasks are scored by DeepSeek (`deepseek-v4-pro`, temperature 0, 4,096-token limit), which accepts equivalent numeric surface forms such as 0.146 and 14.6%, together with equivalent sign and unit conventions. The judge prompt and settings are identical for every configuration, and DeepSeek provides all reported scores on these tasks. Generations that reach the token limit are marked incorrect without judging.

3.5 LLM Judge Selection and Validation

We validate our choice of LLM judge in three steps. First, an independent second judge (Claude Sonnet 5) re-judges 3,000 responses drawn from all judged configurations, blind to DeepSeek's verdicts. The two judges agree on 91.6% of them ($\kappa = 0.76$).

Second, we compare both judges against human labels. An author labels all 252 cases on which the judges disagree, plus 252 sampled cases on which they agree, using written scoring rules and without seeing either judge's verdict or the model configuration. After weighting the two groups by their frequency, DeepSeek agrees with the human on 91.4% of verdicts (Cohen's $\kappa = 0.83$), compared with 90.0% for Claude. Among cases where the judges disagree, the human agrees with DeepSeek three times out of five. The two judges also show different error patterns: DeepSeek tends to reject answers that the human accepts, whereas Claude tends to accept answers that the human rejects. We therefore use DeepSeek as the reporting judge because it is closer to the human labels and less likely to credit borderline answers, which, if anything, makes the reported recovery estimates conservative.

Third, we re-score every configuration with Claude to test whether the choice of judge affects our results. Absolute scores increase by four to fifteen points, but all significant calibration contrasts remain significant and retain the same direction. Our conclusions therefore do not depend on the choice of judge.

3.6 Statistical Analysis

We test each calibration contrast separately for each task. For tasks reported with accuracy, we apply an exact two-sided McNemar test to paired correctness outcomes. For tasks reported with F1, we resample paired predictions over test examples 2,000 times and recompute F1 on each bootstrap sample. Stars indicate unadjusted task-level significance at the thresholds reported in the result tables.

WikiText test perplexity is computed over non-overlapping 2,048-token windows. We also test the sampling stability of the pruning results by repeating SparseGPT and Wanda with three independently drawn WikiText calibration sets. Each run is compared with the same full FinMix corpus.

4 Degradation under Generic Calibration

RQ1 asks what compression damages in financial LLMs and whether the field's standard checks would catch it. Table 3 reports every method under WikiText calibration against the BF16 reference on all ten tasks. The results show two patterns. First, quantization usually causes small losses, pruning causes much larger losses, and numerical QA loses more than classification under every method. Second, perplexity does not reliably show these differences across tasks.

4.1 Task-Selective Damage

For all six methods in Table 3, the average score drops more on numerical QA than on classification. The size of the drop, however, depends strongly on the compression method. Among the quantization methods, SmoothQuant and AWQ leave both task families largely intact, while RTN and GPTQ cause moderate losses that fall more heavily on numerical QA. SparseGPT and Wanda produce much larger losses, and numerical QA again suffers more than classification. In short, numerical QA is more vulnerable under every method, and pruning causes the largest overall decline. We exclude the near-chance stock-movement task (BigData22) from the grouped classification mean, because every model remains close to chance. Score changes mainly reflect different guessing behavior rather than useful prediction skill.

Overlong generation is another visible form of pruning damage. BF16 models typically reach the 2,048-token answer limit on fewer than 4% of QA examples, while the most damaged pruning configuration reaches it on about 45%. Inspection of the limit-hit outputs shows repeated or off-track reasoning without a final answer, rather than valid solutions cut short by the token budget. Reaching the limit therefore reflects a failure to terminate, not simply a need for a longer answer. These outputs remain in the evaluation set and are counted as incorrect.

GPTQ and RTN provide the clearest comparison among the quantization methods. Both use the same four-bit weight format, but only GPTQ uses calibration examples to guide weight reconstruction. GPTQ has better average task performance than RTN, yet this advantage comes mainly from classification. Its numerical-QA loss remains comparable to RTN's. Thus, using calibration examples during reconstruction appears to change which task family bears the loss, not only the total amount of loss.

This comparison cannot establish a mechanism, because GPTQ and RTN differ in more than their use of calibration data. Section 5 therefore holds the compression method fixed and changes only the calibration corpus: if calibration matters for numerical QA, a corpus of financial task examples should help when generic calibration causes substantial losses and matter little when it causes little damage.

4.2 Limits of Perplexity

Perplexity measures how well a language model predicts the next token in a text corpus. Lower values are better. Because it requires no task labels, it is commonly used as a quick quality check after compression. We compute it on held-out WikiText and ask whether models with better perplexity also lose less performance on the financial tasks in Table 3.

For classification, ranking the compressed models by WikiText perplexity gives the same order as ranking them by their average score loss. For numerical QA, it does not. GPTQ, for example, has better perplexity than AWQ but worse performance throughout the numerical-QA block. Better perplexity on generic text therefore does not imply better numerical-QA performance.

A second limitation is that perplexity cannot show which downstream capabilities compression has damaged. It assigns one overall score to each model, while Table 3 shows that the same compressed model can retain classification performance and lose much of its numerical capability. Wanda is the clearest example: it retains much more classification performance while numerical QA largely collapses. A single perplexity value cannot represent this difference.

Answer to RQ1. Under every method tested, numerical QA loses more than classification, and pruning causes the largest absolute losses. Perplexity ranks the methods correctly for classification but not for numerical QA, and one perplexity value cannot show which tasks have deteriorated.

5 Recovery with Domain-Matched Calibration

RQ2 asks when domain-matched calibration data recovers performance lost to compression. It also asks whether the improvement is specific to FinMix or would arise after any change of calibration corpus. We replace WikiText with FinMix at the same token budget, with no retraining and no change to the compression method. Replacing WikiText with C4 provides a generic-corpus control. We use *recovery* for the measured gain in task score and *repair* for the act of correcting damage or an individual error.

5.1 Recovery Beyond a Generic-Corpus Swap

Figure 1 compares the improvement from FinMix with the change produced by switching between generic corpora. The left panel shows the damage under WikiText calibration. The middle and right panels then compare C4 and FinMix, respectively, against the same WikiText baseline. The middle panel reports C4–WikiText, and the right reports FinMix–WikiText. Using the same baseline shows whether broad improvements appear after either corpus

Table 3: Performance of Llama-3.1-8B under WikiText calibration, with BF16 as the reference (RTN is calibration-free). Higher task scores are better. Lower PPL is better. Avg. is the unweighted mean within each task family. The parenthetical Δ gives the damage relative to BF16 (BF16 − compressed), so positive values indicate a loss. The classification mean excludes BigData22. Stars denote significant differences from BF16 ($^{*}p<.05$).

Calibration: WikiText			*Classification*							*Numerical QA*				
	Method	PPL↓	FPB↑	FOMC↑	Headlines↑	FinRED↑	FinCausal↑	BigData22↑	***Avg.***↑	CFA↑	FinQA↑	ConvFinQA↑	TAT-QA↑	***Avg.***↑
	BF16	7.22	79.0	51.3	67.9	75.2	53.5	49.6	*65.4*	68.1	43.9	64.7	74.1	*62.7*
Quantization	RTN	7.96	78.7	50.4	58.1*	73.8	47.4*	43.4*	*61.7 (Δ 3.7)*	62.4*	39.6*	59.1*	69.5*	*57.6 (Δ 5.1)*
	GPTQ	7.49	77.9	51.9	68.0	72.0*	53.8	58.3*	*64.7 (Δ 0.7)*	62.8*	36.9*	58.6*	70.5*	*57.2 (Δ 5.5)*
	AWQ	7.62	76.4*	49.2	68.2	74.0	52.7	49.3	*64.1 (Δ 1.3)*	64.8*	45.8	59.8*	73.5	*61.0 (Δ 1.7)*
	SmoothQuant	7.33	79.0	53.1*	67.7	73.7*	54.3	50.5	*65.6 (Δ −0.2)*	66.8	44.8	63.1	73.0	*61.9 (Δ 0.8)*
Pruning	SparseGPT	9.57	77.7	45.4*	23.7*	67.1*	44.9*	39.9*	*51.8 (Δ 13.6)*	53.5*	24.9*	44.4*	54.7*	*44.4 (Δ 18.3)*
	Wanda	18.17	65.2*	45.0*	41.9*	59.4*	21.1*	43.7*	*46.5 (Δ 18.9)*	32.9*	6.7*	16.6*	31.3*	*21.9 (Δ 40.8)*

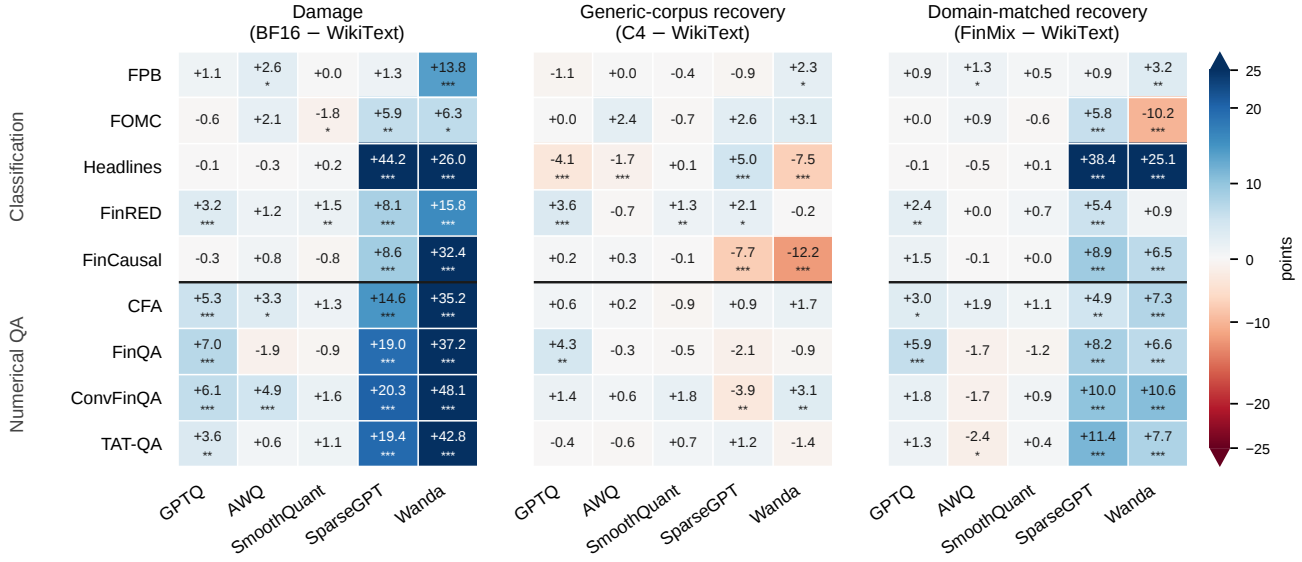


Figure 1: Compression damage under WikiText calibration (left), and the change after replacing WikiText with C4 (middle) or FinMix (right). Positive values indicate damage in the left panel and recovery in the other two panels. Stars indicate paired-test significance ($^{*}p<.05$, $^{**}p<.01$, $^{***}p<.001$).

replacement. Because FinMix changes both financial content and task-style formatting, the comparison measures their joint effect rather than a pure domain effect.

SparseGPT and Wanda show the clearest difference between the two corpus changes. Replacing WikiText with C4 produces mixed changes with no consistent improvement, including some large and significant losses (e.g., −7.7 on FinCausal under SparseGPT and −12.2 under Wanda). In contrast, replacing WikiText with FinMix improves most tasks under both pruning methods, showing a broad improvement that does not appear when moving from one generic corpus to another. It also reduces overlong generation by more than half in the configurations where this pruning failure is most severe.

The other methods behave differently. GPTQ suffers less initial damage than pruning, and neither corpus replacement changes its average performance by much. SmoothQuant and AWQ also change little, consistent with their small losses under WikiText calibration. Thus, FinMix is not uniformly beneficial. Its clearest advantage appears after severe pruning damage.

Overall, FinMix produces more consistent recovery after pruning than the generic-corpus control, but its effect depends on the compression method. We next test whether the observed recovery depends on the particular calibration examples sampled.

5.2 Robustness Across WikiText Draws

To check sampling stability, we repeat SparseGPT and Wanda with three independently sampled WikiText calibration sets and compare each result with the same full FinMix corpus. The check covers three classification and three numerical-QA tasks.

Table 4: Pruning recovery across three WikiText draws. Entries give significantly positive comparisons over all task–seed comparisons ($p<.05$). All 36 recovery values are positive.

Method	*Classification*	*Numerical QA*	Overall
SparseGPT	8/9	9/9	17/18
Wanda	9/9	9/9	18/18

Table 5: FinMix repair rates among errors under WikiText calibration, grouped by whether BF16 answers the same sample correctly. The higher rate for BF16-correct samples indicates recovery of performance lost during compression.

	FinMix repair rate (%)		
Method	BF16 correct	BF16 wrong	OR [95% CI]
GPTQ	62.1	15.2	9.08 [6.95, 11.86]
SparseGPT	48.0	16.1	4.62 [3.69, 5.80]
Overall	53.2	15.6	6.07 [5.11, 7.22]

OR denotes the odds ratio. The GPTQ and SparseGPT estimates are stratified by task. The Overall row combines both methods and is stratified by method and task.

Table 4 shows that recovery is positive in all 36 method–task–seed comparisons and significant in 35. Every numerical-QA comparison is significantly positive for both pruning methods. The only nonsignificant result is one SparseGPT classification run. The pruning recovery is therefore stable across different WikiText calibration samples, with the strongest consistency on numerical QA.

5.3 Which Errors Are Repaired

To determine whether FinMix restores performance lost during compression, we examine every error made by GPTQ and SparseGPT under WikiText calibration on FinQA and ConvFinQA. We divide these errors into two groups. In the first, BF16 answers the sample correctly but the compressed model does not. These errors were introduced by compression. In the second, both models answer incorrectly. These errors were already present in BF16.

Table 5 shows the same pattern for both methods. FinMix repairs about half of the errors introduced by compression, but only about one in six of the errors already present in BF16. Thus, an error is far more likely to be repaired when BF16 originally answered the sample correctly. Combining both methods gives an odds ratio of 6.07, with a 95% confidence interval of [5.11, 7.22]. This means that the odds of repair are about six times higher for errors introduced by compression than for errors already made by BF16. FinMix can occasionally solve a sample that BF16 misses, but it is much more effective at recovering performance lost during compression.

We also manually assigned each compression-introduced error to one of six types (reasoning, unit/scale, arithmetic, extraction, format, and other). Reasoning and arithmetic errors together account for about two thirds of them: GPTQ produces relatively more reasoning and unit-or-scale errors, while SparseGPT produces relatively more arithmetic and extraction errors. Across the four common types, GPTQ has higher repair rates than SparseGPT, matching Table 5, but within either method repair rates do not differ reliably across error types (GPTQ $p = 0.50$, SparseGPT $p = 0.086$). The high rates for the rare format and other categories rest on very few cases. Error type thus describes what compression tends to break, but does not predict which errors FinMix repairs.

Answer to RQ2. FinMix provides more consistent recovery than a second generic corpus after severe pruning damage. Its effect is modest for methods with less initial damage. Errors introduced by compression are far more likely to be repaired than errors already present in BF16. Among those compression-introduced errors, the manually assigned error type does not reliably predict repair.

6 Generalization Across Model Families and Scales

RQ3 asks whether the link between compression damage and recovery is limited to Llama-3.1-8B. We test two kinds of generalization. Qwen3-8B changes the model family while keeping a similar scale. Llama-3.1-70B changes the scale while keeping the Llama family.

For each method and task, we define *Damage* (D) as BF16−WikiText and *Recovery* (R) as FinMix−WikiText. Positive D means that compression lowers the score under WikiText calibration. Positive R means that switching to FinMix raises it. Figure 2 plots D against R. Each panel contains 45 points from five methods and nine above-chance tasks. Values are in percentage points, and colors identify compression methods.

The black dashed line is estimated only from panel (a). Its shaded band shows where new results would usually fall if they follow the same relationship. The gray dotted line is full repair, where the gain from FinMix equals all performance lost under WikiText. Points below this line recover only part of the loss. The near-chance BigData22 task is excluded for the reason discussed in Section 4.1.

6.1 Generalization to Qwen3-8B

Figure 2(a) uses only Llama-3.1-8B. Its fitted slope is 0.38, so each additional point of damage is associated with about 0.38 additional points of recovery. Methods with little damage remain near the origin, while pruning produces both larger losses and larger gains from FinMix. The most severely damaged Wanda points remain well below the full-repair line. FinMix therefore recovers only part of the performance lost under extreme pruning.

Figure 2(b) changes the model family but keeps the 8B scale. Its Qwen3-8B points are not used to estimate the line or prediction band. Quantized Qwen3-8B remains near the origin, with little damage and little response to corpus choice. Pruning moves the points outward. The annotated SparseGPT result illustrates the pattern. FinQA suffers substantial damage under WikiText calibration and recovers strongly under FinMix. All 45 Qwen3-8B points fall within the Llama-3.1-8B prediction band.

Across panels (a) and (b), damage and recovery have a Spearman correlation of $\rho = 0.74$ over 90 points. Including the near-chance stock-movement task gives $\rho = 0.68$ over all 100 points. The positive relationship therefore does not depend on that exclusion, and the pattern fitted on Llama-3.1-8B generalizes to a second model family.

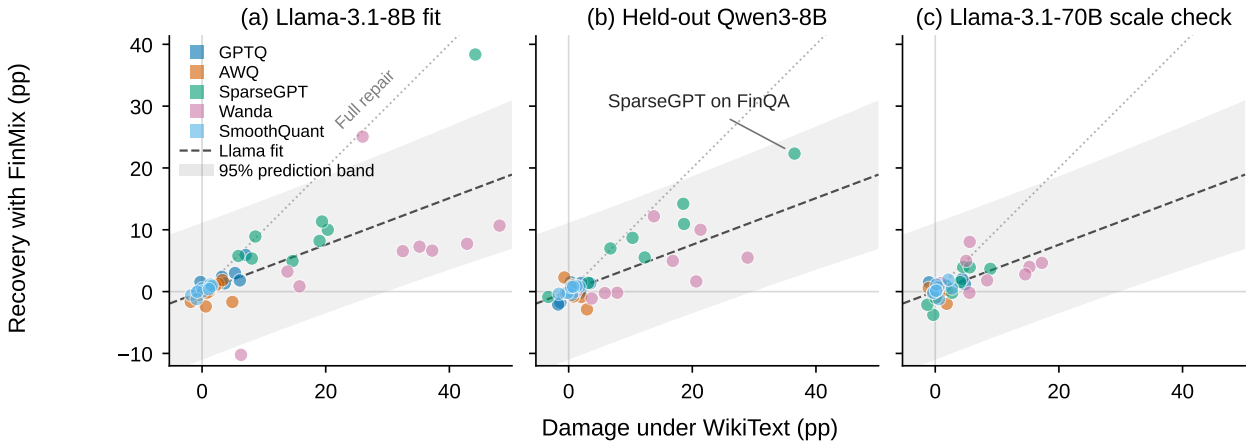


Figure 2: Compression damage and recovery across model families and scales. Panel (a) fits Llama-3.1-8B, panel (b) tests held-out Qwen3-8B, and panel (c) checks Llama-3.1-70B. The black dashed line and shaded area show the fit and its 95% prediction band. The gray dotted line marks full repair.

Table 6: Average BF16 performance and compression damage/recovery (D / R, pp) by method and model. Higher BF16 scores are better.

Method	Llama-3.1-8B	Qwen3-8B	Llama-3.1-70B
Classification			
BF16 score	65.4	66.8	70.7
GPTQ	+0.7/+0.9	−0.3/−0.4	+0.8/+0.6
AWQ	+1.3/+0.3	−0.1/+0.3	+0.2/−0.5
SmoothQuant	−0.2/+0.1	+0.1/0.0	+0.6/+0.2
SparseGPT	+13.6/+11.9	+3.5/+3.3	+1.0/−0.4
Wanda	+18.9/+5.1	+6.5/+2.1	+5.5/+3.8
Numerical QA			
BF16 score	62.7	74.7	77.3
GPTQ	+5.5/+3.0	+2.3/+0.7	+1.9/+1.0
AWQ	+1.7/−1.0	+1.6/−0.9	+0.8/−0.3
SmoothQuant	+0.8/+0.3	+0.7/+0.5	+0.4/+0.1
SparseGPT	+18.3/+8.6	+21.5/+13.3	+5.3/+2.2
Wanda	+40.8/+8.1	+22.0/+5.6	+11.4/+2.3

6.2 Effect of Model Scale

In BF16, classification varies only modestly across models, whereas numerical QA separates them clearly: Qwen3-8B (74.7) is far above Llama-3.1-8B (62.7) and close to Llama-3.1-70B (77.3), so numerical performance depends on model family as well as scale.

The classification block is stable under quantization. GPTQ, AWQ, and SmoothQuant remain close to BF16 on all three models, leaving little for FinMix to recover. The larger differences come from pruning. SparseGPT and Wanda cause substantial classification damage on Llama-3.1-8B, but less on Qwen3-8B and Llama-3.1-70B. Recovery also depends on the pruning method. FinMix removes most of the SparseGPT loss at 8B, whereas the recovery from Wanda is more limited.

Numerical QA shows the clearest scale effect. Quantization remains fairly stable, apart from the larger GPTQ loss on Llama-3.1-8B. Under pruning, Llama-3.1-70B loses much less than either 8B model. The same reduction appears for both SparseGPT and Wanda, suggesting that the larger model is more tolerant of pruning. One possible explanation is greater parameter redundancy at 70B, although our experiments do not test that mechanism directly. Qwen3-8B shows that a strong BF16 score does not provide the same protection. Its numerical performance is close to Llama-3.1-70B before compression, yet it suffers much larger pruning losses, including the largest SparseGPT loss. FinMix recovers part of these losses, but does not fully restore the pruned models.

Panel (c) in Figure 2 shows the same comparison task by task. All 45 points fall within the Llama-3.1-8B prediction band. Most are closer to the origin than the 8B pruning points because compression causes less damage at 70B. With less damage, changing the calibration corpus also produces a smaller gain.

Answer to RQ3. Damage under generic calibration remains a useful indicator of the benefit of FinMix across model families and scales. The Llama-3.1-8B fit generalizes to held-out Qwen3-8B, and the 70B results follow the same direction with smaller damage and smaller recovery. The relationship generalizes, but its magnitude depends on the model and scale.

7 Practical Guidance

Our findings suggest a three-step procedure for deploying compressed models in financial NLP.

(1) **Compare the compressed model directly with BF16.** Run both models on the same representative validation examples and report classification and numerical QA separately. If the application requires calculations or reasoning over tables, include such examples in this check. A combined score or generic perplexity can hide the numerical damage.

(2) **Choose the response based on the observed score loss.** If the loss is already small, changing the calibration corpus

is unlikely to help much. If pruning causes a large loss, recalibrate with task-formatted financial data. Replacing one generic corpus with another did not provide the same reliable recovery. If the remaining loss is still unacceptable, reduce the pruning level or choose a less damaging compression method.

(3) **Repeat the comparison after changing the model or scale.** Qwen3-8B approaches the 70B model in uncompressed numerical performance but suffers much larger pruning losses. A strong BF16 score therefore cannot replace a compression test on the new model. Use the newly measured damage to judge whether changing the calibration data is likely to help.

8 Conclusion

This work reframes calibration-data selection as a response to compression damage rather than a fixed preprocessing choice. The question is not whether domain-matched calibration is always better. It is whether compression has degraded capabilities that better calibration data can recover. When little has been lost, changing the corpus has little value. When a capability has been severely damaged, task-formatted calibration can recover part of the loss. This distinction matters for financial LLMs in particular, because aggregate scores and generic language-model metrics can hide losses in numerical reasoning. Calibration should therefore follow a task-level assessment of what compression has changed.

Our study has limitations. FinMix changes both financial content and task format, so the two effects are not separated. The scale analysis uses a single 70B model, and the mechanism behind its smaller losses is not tested directly. Whether the damage-first rule extends to other specialized domains remains an open question. Within the settings tested, however, measured task-level damage provides a clearer basis than domain membership for deciding when specialized calibration data is worth the effort.

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